Biol Cybern (2010) 103:387–400
DOI 10.1007/s00422-010-0406-6

ORIGINAL PAPER

Biological Cybernetics

Extended causal modeling to assess Partial Directed Coherence in multiple time series with significant instantaneous interactions

Luca Faes · Giandomenico Nollo

Received: 20 July 2009 / Accepted: 22 September 2010 / Published online: 12 October 2010
© Springer-Verlag 2010

Abstract The Partial Directed Coherence (PDC) and its generalized formulation (gPDC) are popular tools for investigating, in the frequency domain, the concept of Granger causality among multivariate (MV) time series. PDC and gPDC are formalized in terms of the coefficients of an MV autoregressive (MVAR) model which describes only the lagged effects among the time series and forsakes instantaneous effects. However, instantaneous effects are known to affect linear parametric modeling, and are likely to occur in experimental time series. In this study, we investigate the impact on the assessment of frequency domain causality of excluding instantaneous effects from the model underlying PDC evaluation. Moreover, we propose the utilization of an extended MVAR model including both instantaneous and lagged effects. This model is used to assess PDC either in accordance with the definition of Granger causality when considering only lagged effects (iPDC), or with an extended form of causality, when we consider both instantaneous and lagged effects (ePDC). The approach is first evaluated on three theoretical examples of MVAR processes, which show that the presence of instantaneous correlations may produce misleading profiles of PDC and gPDC, while ePDC and iPDC derived from the extended model provide here a correct interpretation of extended and lagged causality. It is then applied to representative examples of cardiorespiratory and EEG MV time series. They suggest that ePDC and iPDC are better interpretable than PDC and gPDC in terms of the known cardiovascular and neural physiologies.

Keywords Partial Directed Coherence · Granger causality · Multivariate autoregressive models · Multivariate time series · EEG · Cardiovascular variability

Abbreviations

DC	Directed Coherence
ECG	ElectroCardioGram
EEG	ElectroEncephaloGram
ePDC	extended Partial Directed Coherence
gPDC	generalized Partial Directed Coherence
HF	High Frequency
HP	Heart Period
iPDC	Partial Directed Coherence measured from MVAR model with instantaneous effects
LF	Low Frequency
MVAR	MultiVariate AutoRegressive
PDC	Partial Directed Coherence
RF	Respiratory Flow
SAP	Systolic Arterial Pressure

Electronic supplementary material The online version of this article (doi:10.1007/s00422-010-0406-6) contains supplementary material, which is available to authorized users.

L. Faes (✉) · G. Nollo
Lab. Biosegnali, Department of Physics & BIOTech,
University of Trento, via delle Regole 101,
38123 Mattarello, Trento, Italy
e-mail: luca.faes@unitn.it

G. Nollo
FBK, Via Sommarive 18, 38050 Povo, Trento, Italy

1 Introduction

The extraction of the flow of information from networks of coupled dynamical systems is a topic of great interest in the study of physical processes. Granger causality (Granger 1969), a concept originated in econometrics but rapidly spread out to other fields of science and engineering, has

emerged as a leading tool for exploring this topic. In biomedical applications, evaluation of causality is commonly performed looking for directional dependencies within a set of multiple time series measured from the considered physiological system. In recent years, this approach has become very popular in a number of applications ranging from neurophysiology to cardiovascular physiology (see, e.g., Pereda et al. 2005; Porta et al. 2009 for recent reviews).

Granger causality is formalized in the context of a parametric representation of multiple time series. Specifically, the available data set is regarded as a realization of a linear multivariate stochastic process, and the amount of information flow from one time series to another is evaluated as the improvement in the prediction of the second series brought by consideration of the first series in the parametric model. Although in its original formulation (Granger 1969) and in a number of subsequent developments (Geweke 1982; Bernasconi and Konig 1999; Gourevitch et al. 2006) causality was evaluated in the time domain, the linear parametric representation favors the spectral representation of the time series and of their interaction. This aspect is of particular relevance for the study of physiological processes that are rich of oscillatory content. For this reason, a number of methods, all based on multivariate autoregressive (MVAR) modeling, have been utilized to quantify causality in the frequency domain (Astolfi et al. 2006; Baccala et al. 1998; Baccalà and Sameshima 2001; Faes et al. 2004, 2006; Kaminski et al. 2001; Porta et al. 2002; Schelter et al. 2006). Among them, the directed coherence (DC), defined in the context of neurophysiological or cardiovascular signal analysis, respectively, in (Baccala et al. 1998) and (Porta et al. 2002), is of particular interest as it constitutes a factor in the decomposition of the ordinary coherence function that elicits causality from the spectral representation of the MVAR parameters. While in bivariate systems there is univocal correspondence between DC and direct Granger causality, such a correspondence is not straightforward in multivariate systems as the DC measures both direct and indirect (i.e., mediated by other series) causality between two series. To settle this issue, Baccalà and Sameshima (2001) proposed a generalization of the DC that was termed partial DC (PDC) because it constitutes, under certain assumptions, a factor in the decomposition of the partial coherence function, classically known, e.g., from (Koopmans 1974). Thanks to this property, the PDC is able to detect only direct interactions between two time series in a network of interacting structures. An important refinement of this index is the generalized PDC (gPDC) introduced by the same authors (Baccala et al. 2007) to improve the estimation of the causal coupling in the presence of severely unbalanced model residuals variances. The two estimators have been successfully exploited to depict frequency domain causality in a number of real data applications (see, for instance, Astolfi et al. 2006; Sato et al. 2009; Schlogl and Supp 2006; Wehling et al. 2007; Yang et al. 2005).

So far, the parametric model used in all studies computing PDC and gPDC is a strictly causal MVAR model. The strictly causal representation implies that only lagged effects are modeled, while instantaneous (i.e., not lagged) effects among the time series are not described by any model coefficients. Neglecting instantaneous effects in MVAR models implies that any zero-lag correlation among the time series is translated into a correlation among the model residuals (Lutkepohl 1993). The original definitions of PDC and gPDC explicitly forsake the off-diagonal terms of the covariance model residuals matrix, thus omitting the instantaneous correlations among the series from the frequency domain evaluation of causality. This choice was based on the fact that the notion of Granger causality, as it is used in the very many modern applications, involves the exclusive consideration of lagged effects. However, Granger explicitly discussed the role of instantaneous causality in his seminal paper (Granger 1969), and it is known that instantaneous correlations may severely affect the classical, strictly causal parametric representation of multiple time series (see Cox 1981; Lutkepohl 1993). More generally, the cross covariance among the model residuals is central to the development of well-known tests for Granger causality (Pierce and Haugh 1977; Pierce 1979), and several statistical tools exist which relate its significance to the adequacy of the considered model (Li 2004). Despite this, statistical tests on the residual correlation are almost invariably omitted in studies assessing frequency domain causality through PDC/gPDC computation.

In this study we investigate the impact on the evaluation of frequency domain causality of the presence of significant instantaneous correlations among multiple time series. We show that the strictly causal MVAR representation may produce misleading PDC and gPDC profiles when instantaneous effects are not trivial. Further, we propose the utilization of an extended MVAR model combining instantaneous and lagged effects to improve the assessment of causality in the frequency domain. This model allows evaluation of an extended form of causality including instantaneous effects in addition to the lagged ones, but can be adopted also to infer Granger causality through exclusive consideration of lagged effects. The PDC functions derived from the extended model are compared with the traditional PDC and gPDC calculated for theoretical examples reproducing known conditions of interaction, even at zero lag, and, subsequently, for two real multichannel time series derived from cardiovascular and neural systems.[1]

[1] All computations were performed using Matlab®, version 7.01 (Release 14, 2004).

2 Multivariate autoregressive processes and causality definitions

2.1 Strictly causal multivariate autoregressive processes

Let us consider M observed time series of length N, $y_i(n)$, $i = 1, \ldots, M$, $n = 1, \ldots, N$, collected in the vector $\mathbf{Y}(n) = [y_1(n), \ldots, y_M(n)]^{\mathrm{T}}$. In the linear signal processing framework, the concept of causality is defined in terms of multivariate autoregressive (MVAR) processes. Specifically, it is assumed that $\mathbf{Y}(n)$ can be suitably described as a MVAR process

$$\mathbf{Y}(n) = \mathbf{A}(z)\mathbf{Y}(n) + \mathbf{U}(n), \tag{1}$$

where $\mathbf{U}(n) = [u_1(n), \ldots, u_M(n)]^{\mathrm{T}}$ is a vector of zero-mean white noise input processes with covariance matrix $\mathbf{\Sigma_U} = \left\{\sigma_{ij}^2\right\}$ and $\mathbf{A}(z) = \Sigma_{k=1}^{p} \mathbf{A}_k z^{-k}$ is an $M \times M$ matrix of polynomials where z^{-1} represents the one-lag delay operator (i.e., $z^{-1}\mathbf{Y}(n) = \mathbf{Y}(n-1)$). The parameter p is the MVAR model order, defining the maximum delay used to describe linear dependencies of $\mathbf{Y}(n)$ on its own past. The elements of $\mathbf{A}(z)$ on the main diagonal:

$$A_{ii}(z) = a_{ii}(1) z^{-1} + \cdots + a_{ii}(p) z^{-p}, \quad i = 1, \ldots, M, \tag{2}$$

describe the dependence of the current point of the i-th time series, $y_i(n)$, on its own p past points collected in the vector $\bar{Y}_i = [y_i(n-1), \ldots, y_i(n-p)]$, while the off-diagonal elements of $\mathbf{A}(z)$:

$$A_{ij}(z) = a_{ij}(1) z^{-1} + \cdots + a_{ij}(p) z^{-p}, \quad i, j = 1, \ldots, M, \quad i \neq j, \tag{3}$$

describe the dependence of $y_i(n)$ on $\bar{Y}_j = [y_j(n-1), \ldots, y_j(n-p)]$.

The MVAR model (1) is strictly causal, in the sense that it accounts only for the effects of the past of a time series on another series, while instantaneous effects (i.e., effects of $y_j(n)$ on $y_i(n)$) are not described as $\mathbf{A}_0$ is missing. With this model representation any possible zero-lag correlation among the observed series y_i cannot be described by the model coefficients and thus has to be explained by correlations among the inputs u_i. As a result, the noise covariance $\mathbf{\Sigma_U}$ is generally not diagonal.

2.2 Multivariate autoregressive processes with instantaneous effects

As an alternative to using the model in (1), the multivariate process $\mathbf{Y}(n)$ can be described including instantaneous effects into the interactions allowed by the model. This is achieved considering the MVAR process

$$\mathbf{Y}(n) = \mathbf{B}(z)\mathbf{Y}(n) + \mathbf{W}(n), \tag{4}$$

where $\mathbf{W}(n) = [w_1(n), \ldots, w_M(n)]^{\mathrm{T}}$ is a vector of zero-mean uncorrelated white noise processes with diagonal covariance matrix $\mathbf{\Sigma_W} = \mathbf{\Lambda} = \mathrm{diag}\left\{\lambda_{ii}^2\right\}$, and the matrix modeling the interactions is $\mathbf{B}(z) = \Sigma_{k=0}^{p} \mathbf{B}_k z^{-k}$. The difference with respect to strictly causal MVAR modeling as in (1) is that now the lag variable k takes the value 0 as well, which brings instantaneous effects from $y_j(n)$ to $y_i(n)$ into the model in the form of the coefficients $b_{ij}(0)$ of the matrix $\mathbf{B}_0$. Hence, the elements of $\mathbf{B}(z)$ are:

$$B_{ij}(z) = b_{ij}(0) + b_{ij}(1) z^{-1} + \cdots + b_{ij}(p) z^{-p}, \quad i, j = 1, \ldots, M. \tag{5}$$

Note that, with $i \neq j$, $B_{ij}(z)$ describes the dependence of $y_i(n)$ on the present and the p past points of y_j collected in the vector $Y_j = [y_j(n), \bar{Y}_j] = [y_j(n), y_j(n-1), \ldots, y_j(n-p)]$.

In the MVAR model (4), absence of correlation among the noise inputs w_i, i.e., diagonality of $\mathbf{\Sigma_W}$, is guaranteed by the presence of the instantaneous effects. Indeed, the model (1) can be transformed into the model (4) by left-multiplying (1) by the matrix $\mathbf{L}^{-1} = (\mathbf{I} - \mathbf{B}_0)$, where $\mathbf{I}$ is the $M \times M$ identity matrix and $\mathbf{L}$ results from the Cholesky decomposition applied to the covariance of $\mathbf{U}(n)$:

$$\mathbf{\Sigma_U} = \mathbf{L}\Lambda\mathbf{L}^{\mathrm{T}}. \tag{6}$$

It is easy to show that, with the above transformation, we have $\mathbf{B}_k = \mathbf{L}^{-1}\mathbf{A}_k$, $k = 1, \ldots, p$, and the noise covariance of $\mathbf{W}(n) = \mathbf{L}^{-1}\mathbf{U}(n)$ is the diagonal matrix $\mathbf{\Sigma_W} = \mathbf{\Lambda}$. In fact, in the model (1) the input noise covariance $\mathbf{\Sigma_U}$ is not diagonal whenever instantaneous effects among the series are present, as $\mathbf{B}_0 \neq 0$ yields $\mathbf{\Sigma_U} \neq \mathbf{\Lambda}$. Obviously, if the matrix $\mathbf{B}_0$ has all zero entries we have $\mathbf{L} = \mathbf{I}$ and thus the model (4) reduces to (1) ($\mathbf{A}_k = \mathbf{B}_k$, $\mathbf{U}(n) = \mathbf{W}(n)$, $\mathbf{\Sigma_U} = \mathbf{\Lambda}$). By contrast, the presence of instantaneous effects modeled by $\mathbf{B}_0 \neq 0$ makes all coefficients $\mathbf{B}_k$ (not only $\mathbf{B}_0$) different from $\mathbf{A}_k$.

As a result of the Cholesky decomposition, $\mathbf{L}$ is a lower triangular matrix and $\mathbf{B}_0$ is also lower triangular with null diagonal. To fulfill this constraint in practical applications, the observed time series have to be ordered in a way such that, for each $j < i$, instantaneous effects are allowed from $y_j(n)$ to $y_i(n)$ ($b_{ij}(0) \neq 0$) but not from $y_i(n)$ to $y_j(n)$ ($b_{ji}(0) = 0$). As explained in (Baselli et al. 1997), to make the model identifiable the instantaneous transfer paths have to be imposed, avoiding zero-delay loops in accordance with the canonical form set for the MVAR process. When a priori knowledge of the system is available, it may be used to guide the decision about which instantaneous effects have to be allowed; for instance, if $y_j(n)$ is measured in real time before $y_i(n)$,

then y_j has to precede y_i in the ordering of the series ($j < i$) so that $b_{ij}(0) \neq 0$ and $b_{ji}(0) = 0$.

2.3 Causality definitions for MVAR processes

According to the Granger's definition of causality (Granger 1969), a time series is called causal to another time series if the knowledge of the past of the first series significantly improves prediction of the second series. With the notation introduced in Sect. 2.1, we say that y_j causes y_i if $\bar{Y}_j$ is useful to predict $y_i(n)$. Since this form of causality is based on the exclusive consideration of lagged effects from one series to another, we will denote it as *lagged causality*. Another form of causality is the so-called *instantaneous causality*, which occurs from y_j to y_i if $y_j(n)$ is useful to predict $y_i(n)$. This concept, which was also explicitly defined by Granger (1969), refers to the presence of zero-lag correlations between y_i and y_j, so that the described influence is instantaneous in time (see, also, Lutkepohl 1993; Cox 1981). The concepts of instantaneous and lagged causality can be combined together to formulate the following definition of *extended causality*: the series y_j is causal to the series y_i if $Y_j = [y_j(n), \bar{Y}_j]$ is useful to predict $y_i(n)$.

The definitions above can be directly formulated in terms of the MVAR representation of multiple time series provided by the models defined in (1) and (4). To this end, the presence of causal interactions between two time series is related to the presence of nonzero off-diagonal elements of the coefficient matrices. We note that the strictly causal MVAR representation (1) allows to define only lagged causality, while the extended representation (4) allows inference of all the three forms of causality above defined. In terms of the MVAR coefficients, lagged causality from y_j to y_i is detected for the model (1) or for the model (4) when at least one of the coefficients $a_{ij}(k)$ in (3), or at least one of the coefficients $b_{ij}(k)$ in (5), respectively, is significantly different from zero ($k = 1, \ldots, p$). Moreover, instantaneous causality between y_i and y_j is detected when one of the coefficients $b_{ij}(0)$ and $b_{ji}(0)$ is significantly different from zero, while extended causality from y_j to y_i occurs when at least one $b_{ij}(k)$ in (5), $k = 0, 1, \ldots, p$, is significant. Thus, the definition of extended causality incorporates both instantaneous ($k = 0$) and lagged ($k > 0$) effects. We remark that, although lagged causality is defined with complete analogy for the representations in (1) and (4), different lagged causality patterns may be found depending on whether the strictly causal model or the extended model is used to fit the available multiple time series. Indeed, while in the case of $\mathbf{B}_0 = 0$ instantaneous causality is missing and extended causality is identical to lagged causality, when $\mathbf{B}_0$ is significant not only extended causality differs from lagged causality, but also lagged causality estimated from the model (4) differs from lagged causality estimated from the model (1) because the inclusion of instantaneous effects in the MVAR model changes also the values of the lagged effects ($\mathbf{B}_k \neq \mathbf{A}_k$ for $k > 0$).

3 Extension of the concept of Partial Directed Coherence

3.1 Partial Directed Coherence and generalized Partial Directed Coherence

The Partial Directed Coherence (PDC) (Baccalà and Sameshima 2001) is a frequency domain measure of the directed influences between pairs of signals in a multivariate data set. The classical PDC definition is based on describing the observed multivariate process $\mathbf{Y}(n)$ by means of a strictly causal MVAR model as in (1). Specifically, the PDC from y_j to y_i is defined as

$$\pi_{ij}(z) = \frac{\bar{A}_{ij}(z)}{\sqrt{\sum_{m=1}^{M} \left|\bar{A}_{mj}(z)\right|^2}} \tag{7}$$

where $\bar{A}_{ij}(z) = \delta_{ij} - A_{ij}(z)$ with $\delta_{ij} = 1$ for $i = j$ and $\delta_{ij} = 0$ for $i \neq j$. Moving from the z domain to the Fourier domain, the PDC as function of frequency is obtained by evaluating $\pi_{ij}(z)$ on the unit circle of the complex plane: $\pi_{ij}(f) = \pi_{ij}(z)|_{z=e^{2\sqrt{-1}\pi f/f_s}}$, where f_s is the sampling frequency. The normalization properties $0 \leq \left|\pi_{ij}(f)\right|^2 \leq 1$ and $\Sigma_{i=1}^{M} \left|\pi_{ij}(f)\right|^2 = 1$ indicate that the squared magnitude of the PDC function $\pi_{ij}(f)$ quantifies the strength of the causal coupling from y_j to y_i as a function of frequency, being 0 when y_j does not cause y_i at frequency f, and 1 when all causal influences originating from y_j at frequency f are directed toward y_i. The main peculiarity of PDC is in the fact that it quantifies exclusively the interactions from y_j to y_i that cannot be explained by any other observed time series. Thanks to this property, the PDC estimator can be regarded as an alternative frequency domain representation of the concept of Granger causality in multivariate time series analysis. In fact, the definition of lagged causality given for the MVAR process (1) can be formulated in the frequency domain using the PDC function, stating that, with $i \neq j$, y_j causes y_i at the frequency f if $\bar{A}_{ij}(f) = -A_{ij}(z)|_{z=e^{2\sqrt{-1}\pi f/f_s}}$, and thus $\pi_{ij}(f)$, differs significantly from zero.

The generalized PDC (gPDC) was introduced in (Baccala et al. 2007) to achieve a variance stabilization of the frequency domain representation of lagged causality. It is defined as

$$\tilde{\pi}_{ij}(z) = \frac{\frac{1}{\sigma_{ii}} \bar{A}_{ij}(z)}{\sqrt{\sum_{m=1}^{M} \frac{1}{\sigma_{mm}^2} \left|\bar{A}_{mj}(z)\right|^2}}, \tag{8}$$

where σ_{ii}^2 is the variance of the i-th input process u_i. With respect to the original definition, the gPDC satisfies the desirable property of scale invariance, which prevents the coupling measure from being affected by possible unbalancing of the input process variances, thus guaranteeing compensation for the static gain that might affect the measured variables. Exactly as the PDC, the gPDC is a measure of lagged causality, since it is based on strictly causal MVAR model representation and on neglecting the instantaneous correlations among the input processes (only the diagonal terms of $\mathbf{\Sigma_U}$, σ_{ii}^2, appear in (8)).

3.2 Partial Directed Coherence in the presence of instantaneous effects

By analogy with the definitions of PDC and gPDC formulated for a strictly causal MVAR model, we provide now a definition for *extended* PDC (ePDC) that is valid for MVAR models with instantaneous effects in the form of (4). The ePDC from y_j to y_i is defined as

$$\chi_{ij}(z) = \frac{\frac{1}{\lambda_{ii}} \bar{B}_{ij}(z)}{\sqrt{\sum_{m=1}^{M} \frac{1}{\lambda_{mm}^2} \left|\bar{B}_{mj}(z)\right|^2}} \tag{9}$$

where λ_{ii}^2 is the variance of the noise w_i in (4) and $\bar{B}_{ij}(z) = \delta_{ij} - B_{ij}(z)$. As for the PDC and the gPDC, ePDC is expressed in the frequency domain as $\chi_{ij}(f) = \chi_{ij}(z)|_{z=e^{2\sqrt{-1}\pi f/f_s}}$. As outlined in the Appendix, the ePDC defined in (9) can be regarded as a factor in the decomposition of the partial coherence function.

The ePDC provides a frequency domain representation of the concept of extended causality for the model (4), as significant values of $\chi_{ij}(f)$ occur—with $i \neq j$—when the function $\bar{B}_{ij}(f) = -B_{ij}(z)|_{z=e^{2\sqrt{-1}\pi f/f_s}}$, that incorporates both $b_{ij}(0)$ and $b_{ij}(k)$ with $k \geq 1$, is significantly different from zero. To explore lagged causality in the presence of instantaneous effects, we have to exclude $b_{ij}(0)$ from the spectral causality measure. Hence, we set

$$\tilde{B}_{ij}(z) = \bar{B}_{ij}(z) + b_{ij}(0) = \delta_{ij} - \sum_{k=1}^{p} b_{ij}(k) z^{-k} \tag{10}$$

and then define a new PDC function, thereinafter denoted as iPDC to indicate that it is derived from the model with instantaneous effects (4):

$$\tilde{\chi}_{ij}(z) = \frac{\frac{1}{\lambda_{ii}} \tilde{B}_{ij}(z)}{\sqrt{\sum_{m=1}^{M} \frac{1}{\lambda_{mm}^2} \left|\tilde{B}_{mj}(z)\right|^2}}. \tag{11}$$

Hence, ePDC describes extended causality effects in the frequency domain, while PDC/gPDC and iPDC describe lagged causality effects, respectively, when the data are modeled without and with instantaneous effects. The presence of significant instantaneous causality can be evaluated by testing for nonzero values out of the diagonal of the matrix $\mathbf{\Sigma_U}$ in the model (1) or, equivalently, in the matrix $\mathbf{B}_0$ in the model (4). In the absence of instantaneous causality, ePDC and iPDC are equivalent frequency domain causality descriptors of gPDC (and also of PDC if the input processes have all the same variance); on the contrary, significant instantaneous causality makes ePDC and even iPDC different from PDC/gPDC, as the presence of $\mathbf{B}_0$ leads to different values in the functions $\bar{A}_{ij}(z)$, $\bar{B}_{ij}(z)$, and $\tilde{B}_{ij}(z)$.

4 Theoretical examples

In this section we compare the behavior of the PDC functions defined above using simulated MVAR processes with imposed connectivity patterns including instantaneous interactions. In all simulations, autonomous oscillations are obtained for the scalar process y_i representing it as a second-order AR process described by two complex-conjugate poles with modulus ρ and phases $\varphi = \pm 2\pi f$; in this way, effects from $y_i(n-k)$ to $y_i(n)$ are set as $b_{ii}(1) = 2\rho\cos(\varphi)$ and $b_{ii}(2) = -\rho^2$. Directional connections at lag k are then obtained from the process y_j to the process y_i setting nonzero values for the coefficient $b_{ij}(k)$.

Example 1 In the first simulation we consider an MVAR process of order $p = 2$ with $M = 2$ channels, generated by the equations:

$$\begin{cases} y_1(n) = -\rho^2 y_1(n-2) + w_1(n) \\ y_2(n) = y_1(n) - \rho^2 y_2(n-2) + w_2(n) \end{cases}, \tag{12}$$

where w_1 and w_2 are uncorrelated white noise processes with zero mean and unit variance: $w_1, w_2 \equiv \mathrm{WN}(0, 1)$. The modulus and phase of the poles of the two scalar processes are $\rho = 0.95$ and $\varphi = \pi/2$ (frequency $f = 0.25$). The bivariate process (12) is suitably described by the MVAR model with instantaneous effects of Fig. 1a, where influences on the two series coming from their own past are modeled by the coefficients $b_{11}(2) = b_{22}(2) = -\rho^2$ and the instantaneous effect from y_1 to y_2 is accounted for in the form of the coefficient $b_{21}(0) = 1$. The squared ePDC and iPDC functions calculated using (9) and (11), shown in Fig. 1b, provide the correct frequency domain representation of the simulated process. Indeed, iPDC is uniformly zero in both the causal directions $y_1 \rightarrow y_2$ and $y_2 \rightarrow y_1$ indicating the absence of lagged causality between the two series, while the nonzero ePDC indicates the presence of extended causality, in both cases according to the definition that includes instantaneous interactions. When a strictly causal MVAR model is used to describe the same MVAR process (Fig. 1c), instantaneous influences cannot be accounted for and are shifted into

a correlation among the model inputs: in this case, the off-diagonal elements of $\mathbf{\Sigma_U}$ are indeed $\sigma_{12}^2 = \sigma_{21}^2 = 1$. Moreover, as a consequence of the difference between the matrix coefficient $\mathbf{A}_2$ and the true matrix $\mathbf{B}_2$ describing the process ($\mathbf{A}_2 = \mathbf{L}\mathbf{B}_2$), the instantaneous effect $y_1 \rightarrow y_2$ appears to be lagged ($a_{21}(2) \neq 0$ in Fig. 1c). This erroneous interpretation[2] is reflected by the nonzero profiles of the squared PDC and gPDC in Fig. 1d that can provide a misleading indication of lagged causality from y_1 to y_2.

Example 2 The second simulated example consists of a three-channel MVAR process in which both instantaneous and lagged interactions occur according to the equations:

$$\begin{cases} y_1(n) = 2\rho_1 \cos(2\pi f_1)\, y_1(n-1) - \rho_1^2 y_1(n-2) \\ \qquad\qquad + w_1(n) \\ y_2(n) = y_1(n) - \rho_2^2 y_2(n-2) + w_2(n) \\ y_3(n) = 0.8 y_2(n) + 0.2 y_2(n-1) + w_3(n) \end{cases}, \quad (13)$$

with $\rho_1 = 0.9$, $f_1 = 0.1$, $\rho_2 = 0.95 (f_2 = 0.25)$, and $w_1, w_2, w_3 \equiv \mathrm{WN}(0, 1)$. The self-dependencies and causal effects imposed in (13) are consistent with frequency, bandwidth, and interactions commonly observed for cardiovascular and cardiorespiratory variability signals (see Malliani et al. 1991). The model with instantaneous effects of Fig. 2a fully describes the connections set in (13), generating a frequency domain connectivity pattern in which the nonzero ePDC χ_{21} and χ_{32} portray the extended causality over the directions $y_1 \rightarrow y_2$ (instantaneous effect) and $y_2 \rightarrow y_3$ (mixed instantaneous and lagged effect), while the nonzero iPDC $\tilde{\chi}_{32}$ portrays the lagged causality $y_2 \rightarrow y_3$. If we consider the strictly causal MVAR model describing the process (13), we obtain the interconnection structure of Fig. 2c that evidences the presence of spurious effects. Indeed, direct influences at lags 1 and 2 are estimated from y_1 to y_2 in place of the imposed instantaneous interaction, as well as from y_1 to y_3 although no direct effects exist in the original MVAR process for any time lag. As a consequence, both the PDC and the gPDC provide a misleading indication of nonzero frequency domain lagged causality in the directions $y_1 \rightarrow y_2$ (the interaction is instantaneous) and $y_1 \rightarrow y_3$ (no direct interactions exist).

Example 3 In the third simulation, we consider the four-channel MVAR process:

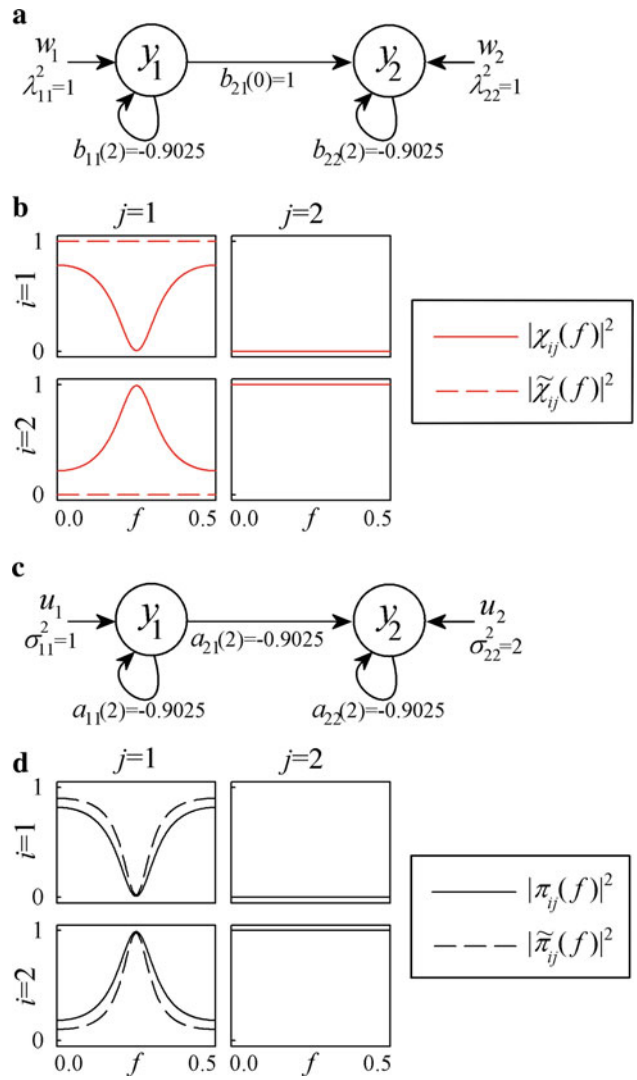

Fig. 1 Representation of the MVAR process described by Eq. (12). Time domain representation through MVAR model with instantaneous effects (**a**) and corresponding frequency domain representation depicting the squared ePDC, $|\chi_{ij}(f)|^2$, and squared iPDC, $|\tilde{\chi}_{ij}(f)|^2$, connectivity estimators (**b**). Time domain representation through strictly causal MVAR model (**c**) and corresponding frequency domain representation depicting the squared PDC, $|\pi_{ij}(f)|^2$, and squared gPDC, $|\tilde{\pi}_{ij}(f)|^2$, connectivity estimators (**d**). When invisible dashed lines corresponding to iPDC (or to gPDC) are superimposed to solid lines corresponding to ePDC (or to PDC)

$$\begin{cases} y_1(n) = 2\rho_1 \cos(2\pi f_1)\, y_1(n-1) - \rho_1^2 y_1(n-2) \\ \qquad\qquad - 0.4 y_3(n-1) + w_1(n) \\ y_2(n) = y_1(n) + 0.2 y_1(n-1) \\ \qquad\qquad - \rho_2^2 y_2(n-2) + w_2(n) \\ y_3(n) = 0.8 y_2(n) + w_3(n) \\ y_4(n) = 0.6 y_2(n) + w_4(n) \end{cases}, \quad (14)$$

with $\rho_1 = 0.95$, $f_1 = 0.125$, $\rho_2 = 0.8$ ($f_2 = 0.25$), and with unbalanced variances for the input processes: $w_1, w_4 \equiv \mathrm{WN}(0, 1)$, $w_2 \equiv \mathrm{WN}(0, 2)$ and $w_3 \equiv \mathrm{WN}(0, 8)$.

[2] It is worth noting that an instantaneous effect may be seen as lagged if the source signal can be described from its own past. This consideration fully holds if the source signal is deterministic, while is valid only with approximation if it is stochastic like the signals modeled here. Nevertheless, it invites to discern some degree of rhetoric in statements such as "erroneous lagged causality" or "spurious effects".

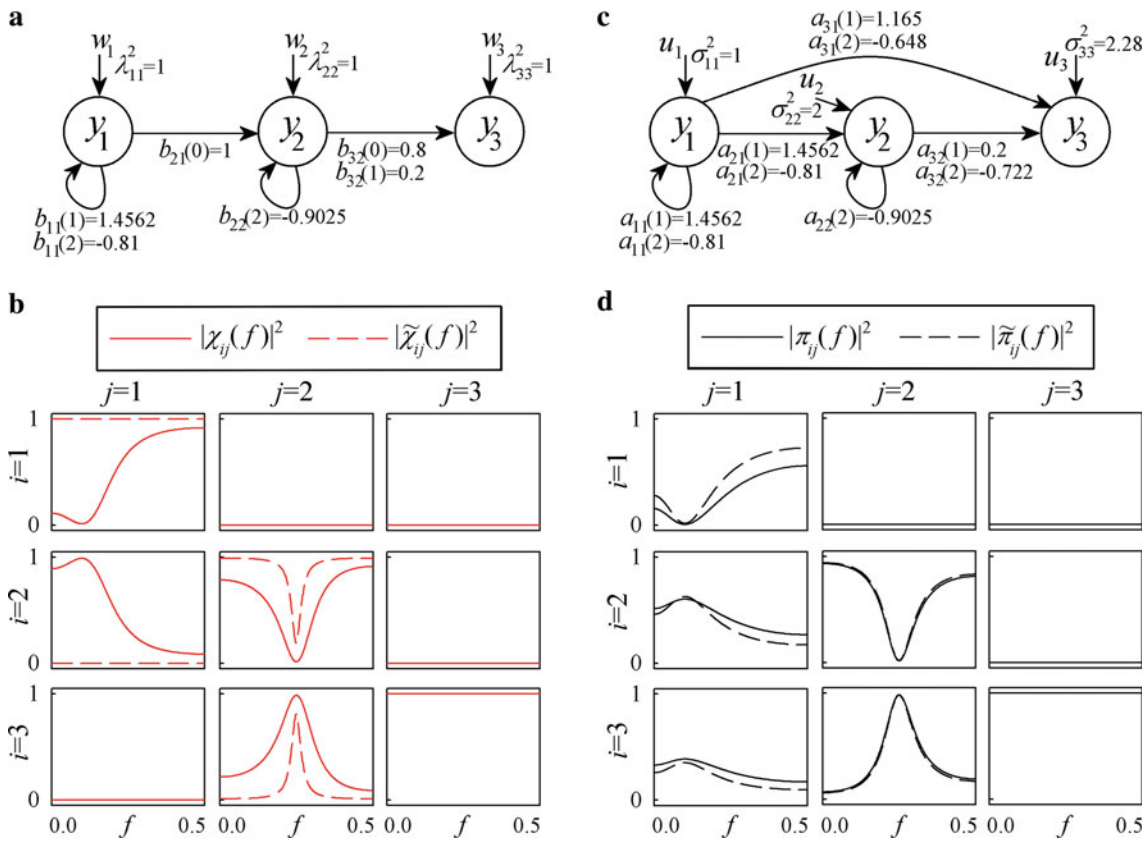

Fig. 2 Representation of the MVAR process described by Eq. (13). Plots and symbols are as in Fig. 1

The parameters set in (14) are consistent with typical EEG activity propagation (see Faes et al. 2009). The imposed interactions are exclusively instantaneous in the directions $y_2 \rightarrow y_3$ and $y_2 \rightarrow y_4$, exclusively lagged in the direction $y_3 \rightarrow y_1$ and mixed instantaneous and lagged in the direction $y_1 \rightarrow y_2$. The utilization of an MVAR model with instantaneous effects results in the structure of Fig. 3a, where the model coefficients are all zero whenever no direct connection is set in (14). The corresponding frequency domain causality descriptors identify the presence of extended causality through the nonzero ePDCs χ_{32}, χ_{42}, χ_{13} and χ_{21}, and of lagged causality through the nonzero iPDCs $\tilde{\chi}_{13}$ and $\tilde{\chi}_{21}$ (Fig. 3b; $\tilde{\chi}_{13}$ is superimposed to χ_{13}). On the contrary, little inference about the correct causal relationships can be made using a strictly causal MVAR model; many spurious lagged effects are indeed estimated both in the derived model structure (Fig. 3c) and in the layout of PDC and gPDC (Fig. 3d). Note that, as a consequence of the normalization condition of the PDC, the presence of spurious causality profiles may lead to an underestimation of the coupling strength over the actual causal direction: e.g., the squared PDC $|\pi_{21}(f)|^2$ remains low due to the nonzero values of the spurious squared PDCs $|\pi_{31}(f)|^2$ and $|\pi_{41}(f)|^2$. Note also that, as a result of the unbalancing of the input variances, the PDC often differs significantly from the gPDC (e.g., over the directions $y_1 \rightarrow y_3$, $y_2 \rightarrow y_3$, $y_3 \rightarrow y_1$); in particular, the erroneously high values of the PDC π_{31} further confound the interpretation of the imposed frequency domain Granger causality pattern.

5 Application to real physiological data

In this Section we report two practical examples concerning the evaluation of causality in the frequency domain using MVAR models with and without instantaneous effects fitted on multivariate physiological time series.[3]

5.1 Experimental protocols

In the first application we consider short-term cardiovascular and cardiorespiratory interactions evaluated in a healthy young subject, breathing spontaneously and standing in the 60° upright position after passive head-up tilt (the example

[3] Files of the analyzed time series are provided as electronic supplementary material (cardiovascular data: Online Resource 1; EEG data: Online Resource 2).

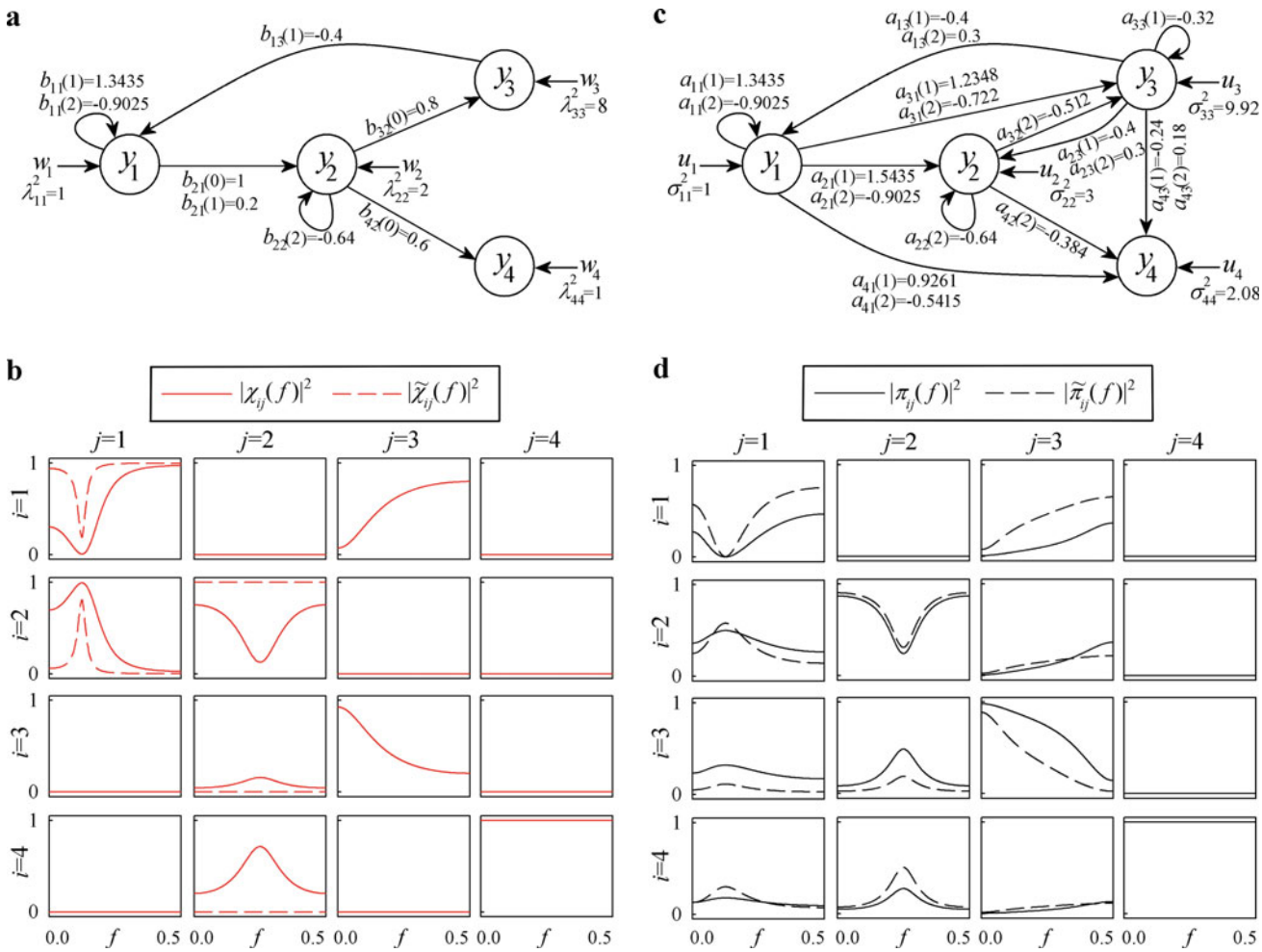

Fig. 3 Representation of the MVAR process described by Eq. (14). Plots and symbols are as in Fig. 1

is taken from Faes et al. 2004). The beat-to-beat variability series of the heart period (HP), systolic arterial pressure (SAP) and respiratory flow (RF) were synchronously measured ($M = 3$), respectively, as the sequences (a) of the temporal intervals between consecutive R peaks of the ECG, (b) of the local maximum of the photopletismographic arterial pressure signal inside each heart period, and (c) of the values of the nasal flow recording, sampled at the onset of the heart period. In accordance with the timings of this measurement convention, instantaneous effects can possibly occur from respiratory flow to systolic pressure and heart period, and from systolic pressure to heart period. Thus, we set $y_1 =$ RF, $y_2 =$ SAP, and $y_3 =$ HP. The synchronous, stationary, and, as far as possible, artifact-free windows of $N = 300$ samples selected for each time series are shown in Fig. 4.

In the second application we considered multichannel ($M = 4$) EEG recordings acquired according to the international 10–20 system (256 Hz sampling rate, with Fpz electrode as common reference) in a healthy young subject resting with eyes closed in the relaxed awake state (data from Erla et al. 2009). Signals were bandpass filtered (FFT filter, 0.3–40 Hz) and then downsampled to 128 Hz. Finally, four bipolar signals taken as representative of different cortical areas were obtained as the difference between pairs of adjacent common-referenced signals (respectively, $y_1 =$ O1−O2: posterior area; $y_2 =$ T3−C3: left central area; $y_3 =$ C4−T4: right central area; $y_5 =$ Fp1−Fp2: frontal area). Stationary windows of 8 s duration ($N = 1024$ samples) were selected for the analysis (see Fig. 5).

5.2 Model implementation and evaluation

For each considered multichannel data set, the coefficients $\mathbf{A}_k$ of the strictly causal MVAR model describing the observed time series as well as the input noise covariance $\mathbf{\Sigma}_\mathbf{U}$, were estimated using the ordinary vector least squares approach (Chatfield 1977; Kay 1988). The coefficients $\mathbf{B}_k$ of the corresponding model with instantaneous effects were then derived as described in Sect. 2.2. The model order was determined following a widely used criterion for MVAR order selection,

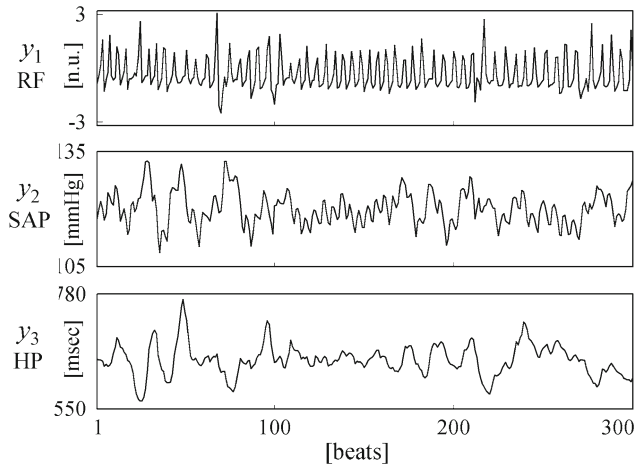

Fig. 4 Time series considered for the cardiovascular variability application: respiratory flow (y_1), systolic arterial pressure (y_2), and heart period (y_3)

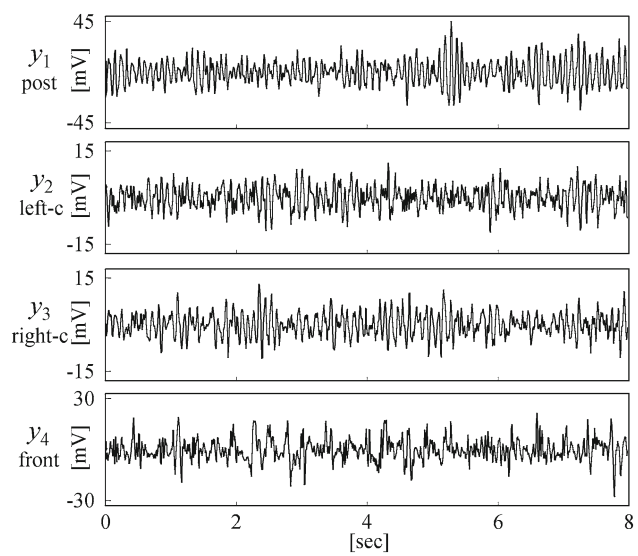

Fig. 5 Time series considered for the neurophysiological application: EEG signals recorded at posterior (y_1), left central (y_2), right central (y_3), and frontal (y_4) head locations

i.e., choosing the order for which the Akaike figure of merit (Akaike 1974), defined as $AIC(p) = N \log\det(\mathbf{\Sigma_U}) + M^2 p$, exhibited a minimum with p in the range (1–30). For the cardiovascular time series, the *AIC* minimum was located at $p = 5$. For the EEG time series, the *AIC* index decreased monotonically with increasing order. As seen in (Brovelli et al. 2004; Ding et al. 2000; Wehling et al. 2007), this appears to be a general feature in studies of the electrical brain activity; in accordance with these studies, we selected the order $p = 5$ as a practical compromise and checked the robustness of the calculated spectral functions against small changes in the model order.

Table 1 Analysis of model residuals, u_i, resulting from strictly causal MVAR modeling of the series y_i, $i = 1, \ldots, M$, for the two practical applications

Cardiovascular

$\mathbf{\Sigma_U}$

RF	0.62	−0.11	0.77
SAP	−0.11	2.34	2.64
HP	0.77	2.64	55.33

$\mathbf{R}$

0.11	−0.09	0.09	
−0.07	*0.24*	**0.23***	
0.07	**0.22***	*0.24*	

EEG

$\mathbf{\Sigma_U}$

post	4.13	0.34	0.49	0.63
left-c	0.34	1.37	−0.41	0.47
right-c	0.49	−0.41	1.02	0.94
front	0.63	0.47	0.94	6.14

$\mathbf{R}$

0.31	0.21*	**0.27***	−0.03
0.12*	*0.48*	**−0.45***	**0.29***
0.22*	**−0.34***	*0.57*	**0.41***
0.11*	**0.15***	**0.34***	*0.44*

Cardiovascular data ($M = 3$, $N = 300$): y_1 = respiration flow, y_2 = systolic arterial pressure, y_3 = heart period; EEG data ($M = 4$, $N = 1024$) : y_1 = posterior EEG, y_2 = left-central EEG, y_3 = right-central EEG, y_4 = frontal EEG. $\mathbf{\Sigma_U}$: estimated residual covariance matrix; **R**: estimated matrix of Spearman rank correlation coefficients between u_i and u_j (lower triangle, $i > j$), Spearman partial correlations between u_i and u_j (upper triangle, $i < j$), and multiple correlations between u_i and all other variables (diagonal, $i = j$). Multiple correlations are printed in italic face. Correlations and partial correlations that are larger than 3 ($N = 300$) or 4 ($N = 1024$) times their estimated standard deviation under the hypothesis of zero true correlation are printed in bold face

*Indicates significant correlation ($p < 0.01$ for $N = 300$ and $p < 0.001$ for $N = 1024$)

To verify the importance of instantaneous effects in the two considered datasets, we evaluated the correlation among the residual time series estimated for the strictly causal MVAR model (1). Results are depicted in Table 1. The estimated residual covariance matrix $\mathbf{\Sigma_U}$ does not exhibit a clear diagonal structure neither for the cardiovascular nor the EEG application. This suggests the presence of correlations among the model residuals in both datasets. To provide statistical validation of this result, we performed a nonparametric Spearman's rank test on the residual time series. The matrix **R** depicted in Table 1 displays both Spearman rank correlation coefficients and Spearman partial correlations between any pair of estimated residuals $u_i(n)$ and $u_j(n)$, $i \neq j$. An estimated correlation or partial correlation is considered to be significantly different from zero if the associated p value is low (see Table). Significant correlation and partial correlation is observed between the residuals u_2 and u_3 for the cardiovascular application, and between all pairs of residuals (except for the partial correlation between u_1 and u_4) for the EEG application. This result can be taken as an indication of the presence of nonzero instantaneous effects among

the observed series. In fact, significant off-diagonal values in $\mathbf{\Sigma_U}$ can be translated into nonzero values in the matrix $\mathbf{B}_0$ modeling instantaneous effects. As a consequence, inference of causality in the frequency domain is expected to lead to different results depending on whether or not instantaneous effects are described by the adopted MVAR model.

Table 2 reports the matrices $|\mathbf{A}|_{\text{mean}}$ and $|\mathbf{B}|_{\text{mean}}$ containing the average magnitude of lagged effects estimated using the strictly causal model and the extended model, respectively.[4] The off-diagonal elements of $|\mathbf{A}|_{\text{mean}}$ and $|\mathbf{B}|_{\text{mean}}$ provide a time domain indication of the strength of the lagged effects exerted along each specific causal direction. Within each matrix, a higher value for the entry (i, j) than for the entry (j, i) suggests the prevalence of the interaction $y_j \rightarrow y_i$ over the interaction $y_i \rightarrow y_j$; a different value for the entry (i, j) of the two matrices suggests a different interpretation of lagged causality due to the influence of instantaneous interactions. According to this interpretation, for the cardiovascular data we observe a prevalence of the directions RF $\rightarrow$ SAP, RF $\rightarrow$ HP, and SAP $\rightarrow$ HP over the opposite directions, as the elements in the lower triangle of $|\mathbf{A}|_{\text{mean}}$ and $|\mathbf{B}|_{\text{mean}}$ have higher magnitude than those in the upper triangle; moreover, the higher value observed for the entry (3,2) of $|\mathbf{B}|_{\text{mean}}$ versus $|\mathbf{A}|_{\text{mean}}$ suggests that a stronger coupling is estimated from SAP to HP using the extended MVAR model. For the EEG application, the strictly causal model does not estimate an evident coupling over some directions of interaction ($|\mathbf{A}|_{\text{mean}}$ values are generally low and comparable); the extended model estimates a much stronger interaction over some directions (e.g., $y_1 \rightarrow y_3$, $y_2 \rightarrow y_3$) and, even more pronounced, ($y_2 \rightarrow y_4$, $y_3 \rightarrow y_4$), leading to evident unidirectional interactions as the strength over the opposite directions remains low. As shown in the next subsection, these results are confirmed, spectrally resolved, by evaluating causality in the frequency domain.

5.3 Frequency domain analysis

Results of the frequency domain analysis performed on the physiological time series are summarized in Figs. 6 and 7. The coefficients of the MVAR model of Eq. 4, estimated as described in Sects. 2.2 and 5.2, were used to compute the power spectral density function of each single time series as $S_{ii}(f) = \sum_{m=1}^{M} \lambda_{mm}^2 |H_{im}(f)|^2$, where $H_{im}(f)$ is the $i - m$ element of the transfer matrix $\mathbf{H}(f) = \left(\mathbf{I} - \mathbf{B}(z)|_{z=e^{2\sqrt{-1}\pi f/f_s}}\right)^{-1}$. Moreover, frequency domain causality was evaluated according to the approaches described in Sect. 3.

[4] Averaging of coefficient magnitudes are chosen to provide an overall, compact representation of time domain effects. The developed Matlab code does not compute coefficient standard errors.

Table 2 Analysis of model coefficients for the two practical applications

Cardiovascular

$|\mathbf{A}|_{\text{mean}}$

$$\begin{bmatrix} 0.22 & 0.04 & 0.01 \\ 0.34 & 0.41 & 0.06 \\ \mathbf{1.53} & 0.24 & 0.47 \end{bmatrix}$$

$|\mathbf{B}|_{\text{mean}}$

$$\begin{bmatrix} 0.19 & 0.03 & 0.01 \\ 0.33 & 0.34 & 0.05 \\ \mathbf{1.50} & \mathbf{0.64} & 0.37 \end{bmatrix}$$

EEG

$|\mathbf{A}|_{\text{mean}}$

$$\begin{bmatrix} \mathbf{1.75} & 0.04 & 0.02 & 0.06 \\ 0.07 & \mathbf{1.41} & 0.12 & 0.02 \\ 0.03 & 0.14 & \mathbf{1.45} & 0.01 \\ 0.07 & 0.20 & 0.05 & \mathbf{1.27} \end{bmatrix}$$

$|\mathbf{B}|_{\text{mean}}$

$$\begin{bmatrix} \mathbf{1.40} & 0.03 & 0.02 & 0.05 \\ 0.08 & \mathbf{1.13} & 0.09 & 0.02 \\ 0.19 & 0.33 & \mathbf{1.13} & 0.01 \\ 0.07 & \mathbf{0.67} & \mathbf{1.60} & \mathbf{1.01} \end{bmatrix}$$

$|\mathbf{A}|_{\text{mean}} = p^{-1} \sum_{k=1}^{p} |\mathbf{A}_k|$ and $|\mathbf{B}|_{\text{mean}} = (p+1)^{-1} \sum_{k=0}^{p} |\mathbf{B}_k|$ quantify the average causal effects estimated for the strictly causal MVAR model and the extended MVAR model, respectively (entries of $|\mathbf{A}|$ are the absolute values of corresponding entries of $\mathbf{A}$). Values larger than 0.5 are printed in bold face. Matrix indices are as in Table 1

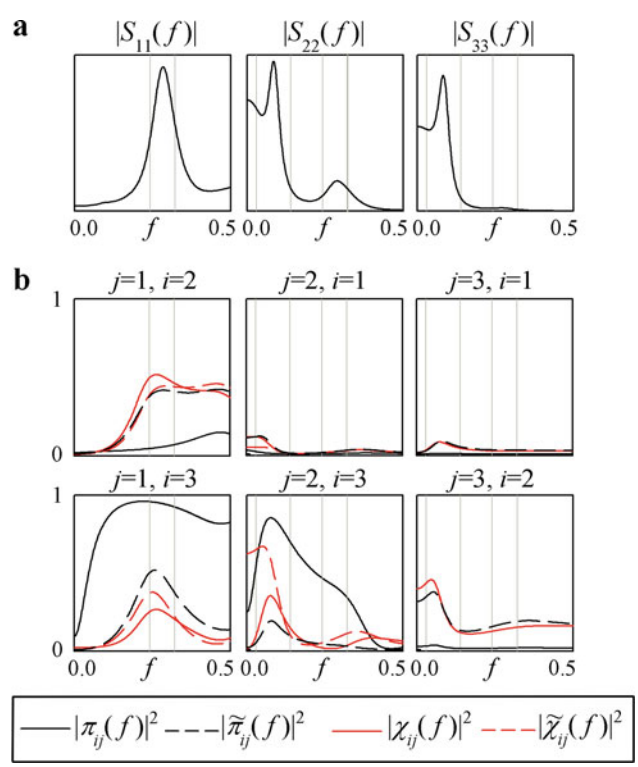

Fig. 6 Frequency domain analysis for the cardiovascular application (y_1: respiratory flow; y_2: systolic arterial pressure; y_3: heart period). Power spectrum of y_i ($S_{ii}(f)$) (**a**) and causal coupling estimated by means of squared PDC, $|\pi_{ij}(f)|^2$, squared gPDC, $|\tilde{\pi}_{ij}(f)|^2$, squared ePDC, $|\chi_{ij}(f)|^2$, and squared iPDC, $|\tilde{\chi}_{ij}(f)|^2$ (**b**). Vertical lines denote the bounds of LF and HF bands of the frequency spectrum

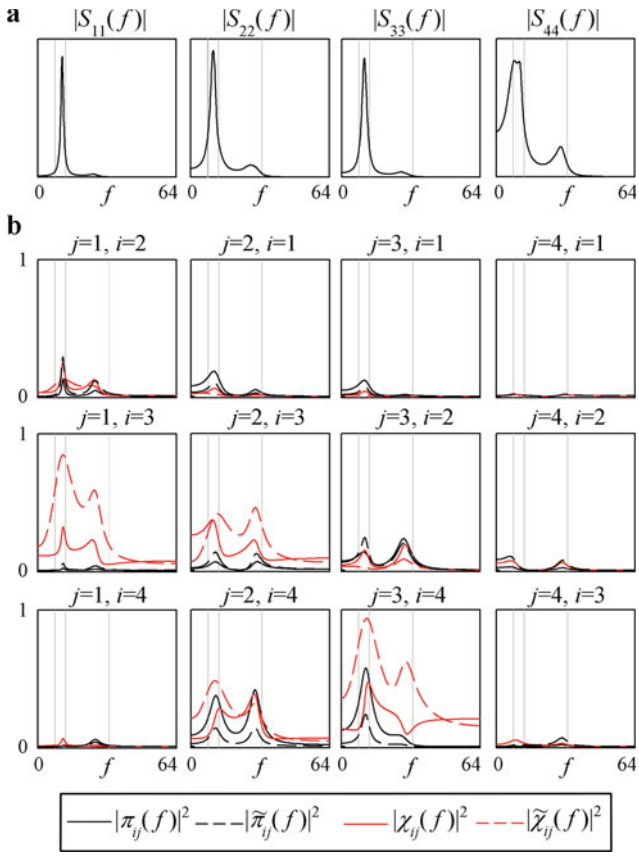

Fig. 7 Frequency domain analysis for the neurophysiological application (y_1: posterior EEG; y_2: left central EEG; y_3: right central EEG; y_4: frontal EEG). Power spectrum of $y_i(S_{ii}(f))$ (**a**) and causal coupling estimated by means of squared PDC, $|\pi_{ij}(f)|^2$, squared gPDC, $|\tilde{\pi}_{ij}(f)|^2$, squared ePDC, $|\chi_{ij}(f)|^2$, and squared iPDC, $|\tilde{\chi}_{ij}(f)|^2$ (**b**). Vertical lines denote the bounds of alpha and beta bands of the frequency spectrum

The power spectra of the cardiovascular time series are shown in Fig. 6a. The RF series exhibits a prominent spectral peak, centered at the respiratory frequency, in the so called high frequency band (HF, around 0.3 Hz in this example); this peak is reflected by synchronous peaks in the SAP and HP series. Moreover, SAP and HP spectra show a major peak in the low frequency band (LF, from 0.04 to 0.15 Hz), which is known to reflect oscillatory behaviors (the so called Mayer waves) not related to respiration (Malliani et al. 1991). According to this scenario, we assessed directional interactions between HP and SAP in the LF band, and between RF and the two cardiovascular variables in the HF band. The different PDC functions (Fig. 6b) provide different directional information about cardiovascular and cardiorespiratory interactions. The squared PDC depicts mostly unidirectional interactions, from SAP to HP (i.e., $|\pi_{32}|^2$) at LF and from RF to HP (i.e., $|\pi_{31}|^2$) at HF; this behavior is very likely misleading as the PDC is affected by the strong unbalancing of the MVAR residual variances, which is clearly evidenced in Table 1 where the diagonal terms of the estimated covariance matrix $\mathbf{\Sigma_U}$ highly differ to each other. The gPDC describes a more balanced interaction, which seems more reliable as it corresponds to known physiological mechanisms: the baroreflex feedback (SAP $\rightarrow$ HP, $|\pi_{32}|^2$) and mechanical feedforward (HP $\rightarrow$ SAP, $|\pi_{23}|^2$) at LF (see Mullen et al. 1997; Nollo et al. 2005), the respiratory sinus arrhythmia (RF $\rightarrow$ HP, $|\pi_{31}|^2$) and the mechanical respiratory driving of arterial pressure (RF $\rightarrow$ SAP, $|\pi_{21}|^2$) at HF (see Baselli et al. 1994; Saul et al. 1989). Compared to gPDC, the ePDC and iPDC functions show superimposable trends in some directions (RF $\rightarrow$ SAP and HP $\rightarrow$ SAP) and different trends in some other directions (RF $\rightarrow$ HP and SAP $\rightarrow$ HP). The differences suggest more reliable values for ePDC/iPDC than for gPDC, as a depressed respiratory sinus arrhythmia (lower RF $\rightarrow$ HP coupling at HF) and an enhanced baroreflex regulation (higher SAP $\rightarrow$ HP coupling at LF) are expected phenomena for subjects standing in the upright position (e.g., observed in (Nollo et al. 2005)).

Figure 7 reports the results obtained for the EEG application. The EEG power spectrum displays for all signals two main peaks in the alpha (8–13 Hz) and beta (13–30 Hz) frequency bands. Confirming expected neurophysiological behaviors (Babiloni et al. 2006), the alpha power is predominant over the beta power in this eyes closed condition, especially in the posterior brain areas. The connectivity pattern represented by the squared PDC and gPDC functions is not clearly indicating a preferential direction of propagation; in particular, the PDC values in both alpha and beta bands are rather low, both from the posterior to the central regions ($y_1 \rightarrow y_2$ and $y_1 \rightarrow y_3$) and from the central to the posterior regions ($y_2 \rightarrow y_1$ and $y_3 \rightarrow y_1$). On the contrary, ePDC and iPDC elicit a clear propagation from the posterior to the right central regions ($y_1 \rightarrow y_3$), while the coupling directed toward the left central regions ($y_1 \rightarrow y_2$) is comparably low with that depicted by PDC/gPDC. Moreover, ePDC and iPDC seem to describe more markedly than PDC/gPDC the directional interactions from the central to the frontal cortical areas ($y_2 \rightarrow y_4$ and $y_3 \rightarrow y_4$), while backward interactions seem to be excluded. These observations suggest that ePDC and iPDC might be proposed as better descriptors than the conventional PDC of an expected neurophysiological behavior, i.e., the propagation of the occipital alpha waves from the posterior to the anterior brain areas (see Kaminski et al. 1995, Ito et al. 2005). Within this picture, the coupling strengths derived using the extended MVAR model indicate a specific propagation pattern going from occipital and left central regions toward right central ($y_1 \rightarrow y_3$ and $y_2 \rightarrow y_3$) and then frontal ($y_2 \rightarrow y_4$ and $y_3 \rightarrow y_4$) regions. Though preliminary, this result agrees with a previous study (Kaminski et al. 1997) suggesting that during eyes closed wakefulness the sources of EEG activity are located in the

left occipital cortex, and propagation occurs from these areas toward right and anterior cortical sites. Finally, we observe higher values of iPDC than ePDC for some connections; in particular, iPDC elicits connectivity from central to frontal areas ($y_2 \rightarrow y_4$, $y_3 \rightarrow y_4$) much better than ePDC, which is in this case comparable to the traditional PDC. This indicates that, for this particular case, a prevalence of lagged causality over extended causality effects in the description of EEG activity propagation.

6 Discussion and conclusions

In its original formulation and in all the numerous subsequent applications, the computation of PDC was based on the identification of a strictly causal MVAR model which does not describe instantaneous effects among the investigated multiple time series. In contrast to this common approach, the present study emphasizes the necessity of including instantaneous effects in the MVAR model used to explore in the frequency domain the concept of causality among multiple time series. One major finding of this study is indeed that, in the presence of nonnegligible instantaneous correlations among the observed time series, the computation of PDC based on the coefficients of a strictly causal MVAR model, either in the original formulation (Baccalà and Sameshima 2001) or in the generalized one (Baccala et al. 2007), may lead to erroneous inferences of frequency domain causality. The reported simulations include examples in which, using the traditional PDC in the presence of zero-lag correlations, instantaneous effects are regarded as lagged, spurious effects appear, and incorrect evaluations of the coupling strength occur. The reason for these misleading interpretations is in the fact that considering instantaneous effects changes the coefficients describing lagged effects as well. Thus, although the idea of forsaking the residual cross-correlations in the definition of PDC may seem reasonable as such correlations are related solely to instantaneous causality (see Lutkepohl 1993), it is not sufficient to provide an unambiguous representation of lagged causality. In the time domain, the information about lagged causality provided by the coefficients of the strictly causal MVAR representation can be extended to instantaneous interactions computing the zero-lag partial correlation of the model residuals (as done in Eichler 2005). However, this additional information cannot be easily integrated into frequency domain causality measures such as the PDC. A possible solution is the one proposed in this study, which consists in shifting the correlation between the residuals into the model coefficients in order to provide a full description of the interactions among the time series; the following PDC computation may then be performed either with adherence to the definition of Granger causality (Granger 1969) which considers only lagged effects, or incorporating also the Granger definition of instantaneous causality so that an extended causality measure is achieved which accounts for both instantaneous and lagged effects. These two ways, leading, respectively, to the definitions of iPDC and ePDC, have been shown to successfully portray frequency domain patterns of lagged causality and extended causality in our simulated examples. The proposed iPDC and ePDC functions were also proven to satisfy, similarly to the gPDC but not the PDC, the property of scale invariance. This aspect is very important in real data applications, as scale invariance assures a reliable inference of the coupling strength when the involved variables are not measured on a common scale.

The reported examples of multivariate physiological time series illustrated the feasibility of extended MVAR modeling to infer frequency domain causality in real data applications. When compared to the traditional PDC/gPDC computed disregarding instantaneous correlations, the proposed ePDC and iPDC functions provided connectivity patterns that were more readily interpretable in accordance with the known cardiovascular and neural physiology. Examples of expected physiological behaviors that were described more clearly by ePDC/iPDC than by PDC/gPDC are the description of cardiovascular and cardiorespiratory mechanisms of modulation of heart period variability, and the detection of patterns of propagation of the alpha EEG rhythms with origin in the posterior brain regions, previously observed in (Nollo et al. 2005) and (Kaminski et al. 1995), respectively. The reported results are certainly preliminary, as they come from single-subject examples. Moreover, issues of spurious correlation or causality arising from different factors (presence of common causes, absence of relevant signals in the considered set, etc.) hold for the proposed as well as for other coupling estimators; for instance, in EEG analysis these issues are related to problems like volume conduction effect and presence of a reference electrode. Nevertheless, the different connectivity pictures provided by ePDC/iPDC on one side, and by the traditional PDC/gPDC on the other side, suggest the importance of accounting for instantaneous correlations to improve the reliability of the frequency domain identification of causality. This point appears crucial in the analysis of physiological time series, also in view of the fact that a significant correlation among the MVAR residuals, indicative of substantial zero-lag interactions, is likely to occur in real data applications; such a correlation was statistically proven in both the real data examples considered in this study.

An issue having implications in the practical application of the proposed approach regards the specifications in the design of the model layout needed to guarantee identifiability of MVAR models with instantaneous effects. Since the signal covariance information alone is not sufficient to define uniquely the model parameters, prior knowledge about the zero-lag structure is usually incorporated into the model definition (Baselli et al. 1997; Korhonen et al. 1996). Specifically,

instantaneous effects can be allowed only in one direction to avoid algebraic loops, and this restriction results in the constraint of acyclicity of the matrix describing the instantaneous effects ($\mathbf{B}_0$ in Sect. 2.2). In the proposed approach this constraint is implicit in the use of the Cholesky decomposition that returns a lower triangular $\mathbf{B}_0$, and is fulfilled by choosing the order of the time series prior to model identification. In real data applications, the required a priori information about the time series should be derived from physical considerations, e.g., based on the temporal order of the events of interest for each measured variable. For instance, in our cardiovascular application the variability series are constructed in a way such that instantaneous effects from heart period to systolic pressure and to respiration, and from systolic pressure to respiration, lack physical meaning and thus can be fixed to zero (of course they are allowed in the opposite directions to explain possible residual correlations). In other situations, like for instance the EEG data analysis, the choice is less straightforward; the evaluation of the best strategy to resolve the resulting arbitrariness is an open issue that goes beyond the scope of this study. Nonetheless, we remark that attempts of estimating models including either instantaneous and/or lagged interactions without prior explicit assumptions are currently under investigation (Shimizu et al. 2006). Methods coming from future work which yield objective estimates of the instantaneous transfer paths could be integrated in the framework proposed in this study for the proper evaluation of PDC in presence of instantaneous correlations occurring in physiological multichannel time series.

In conclusion, the proposed extension of the MVAR model underlying the evaluation of causality in the frequency domain via the PDC function is recommended whenever the time resolution of the measurements is lower than the time scale of the lagged influences describing the observed multivariate process. In such a case, the interpretation of lagged effects may change considerably if instantaneous effects are not included in the model. As reported by our practical examples taken from the study of multiple interactions in cardiovascular physiology and neurophysiology, the significance of instantaneous correlations makes the proposed approach helpful for an adequate inference of causal connectivity patterns among biological rhythms.

Appendix

In this Section we illustrate how the ePDC defined in (9) can be exploited to decompose the partial coherence between two time series taken from a multivariate process consisting of M channels. The partial coherence function $\kappa_{ij}(z)$ (Koopmans 1974) describes the noncausal interaction between the time series y_i and y_j when the influence due to all other M-2 series is discounted. Making use of the factorization suggested in Baccalà and Sameshima (2001):

$$\kappa_{ij}(z) = \frac{\bar{\mathbf{b}}_i^{\mathrm{H}}(z)\,\mathbf{\Lambda}^{-1}\bar{\mathbf{b}}_j(z)}{\sqrt{\left(\bar{\mathbf{b}}_i^{\mathrm{H}}(z)\,\mathbf{\Lambda}^{-1}\bar{\mathbf{b}}_i^{\mathrm{H}}(z)\right)\left(\bar{\mathbf{b}}_j^{\mathrm{H}}(z)\,\mathbf{\Lambda}^{-1}\bar{\mathbf{b}}_j^{\mathrm{H}}(z)\right)}}, \quad \text{(A1)}$$

where $\bar{\mathbf{b}}_i(z)$ represents the i-th column of $(\mathbf{I}\text{-}\mathbf{B}(z))$, $\bar{\mathbf{b}}_i(z) = [\bar{B}_{1i}(z)\cdots\bar{B}_{Mi}(z)]^{\mathrm{T}}$ and the superscript ^{H} stands for hermitian transpose, we have that, with the definition in (8):

$$\kappa_{ij}(z) = \sum_{m=1}^{M} \chi_{mi}^{*}\chi_{mj}. \quad \text{(A2)}$$

Equation A2 shows that ePDC reflects the terms into which the partial coherence can be decomposed. A similar relation is valid for the decomposition of the ordinary coherence function on the basis of the DC evaluated after modeling the instantaneous interactions. Regarding the decomposition shown in (A2), we note that it holds only for MVAR processes with instantaneous effects in which it is assured that the noise covariance matrix $\mathbf{\Lambda}$ is diagonal. On the contrary, a similar factorization cannot be obtained from strictly causal MVAR models since in that case diagonality of the noise covariance matrix $\mathbf{\Sigma_U}$ is not guaranteed. In the particular case of absence of instantaneous interactions among the signals leading to diagonal input noise covariance matrix, the factorization (A2) holds for the gPDC defined in (8).

The ePDC and iPDC functions defined in (9) and (11) share with the gPDC defined in (8) the property of scale invariance, which characterizes also other common measures of linear coupling like the ordinary coherence and the partial coherence, but not the PDC. Moreover, both ePDC and iPDC preserve the normalization properties $0 \leq \left|\chi_{ij}(f)\right|^2 \leq 1$, $0 \leq \left|\tilde{\chi}_{ij}(f)\right|^2 \leq 1$ and $\Sigma_{i=1}^{M}\left|\chi_{ij}(f)\right|^2 = 1$, $\Sigma_{i=1}^{M}\left|\tilde{\chi}_{ij}(f)\right|^2 = 1$ that allow interpretation of the two functions as normalized frequency domain measures of extended causality and of lagged causality ranking the coupling strength with respect to the source of the information flow.

References

Akaike H (1974) A new look at the statistical model identification. IEEE Trans Automat Contr 19:716–723

Astolfi L, Cincotti F, Mattia D, Marciani MG, Baccala LA, De Vico FF, Salinari S, Ursino M, Zavaglia M, Babiloni F (2006) Assessing cortical functional connectivity by partial directed coherence: simulations and application to real data. IEEE Trans Biomed Eng 53:1802–1812

Babiloni C, Binetti G, Cassarino A, Dal Forno G, Del Percio C, Ferreri F, Ferri R, Frisoni G, Galderisi S, Hirata K, Lanuzza B, Miniussi C, Mucci A, Nobili F, Rodriguez G, Luca RG, Rossini PM (2006) Sources of cortical rhythms in adults during physiological aging: a multicentric EEG study. Hum Brain Mapp 27: 162–172

Baccala LA, Sameshima K (2001) Partial directed coherence: a new concept in neural structure determination. Biol Cybern 84: 463–474

Baccala LA, Sameshima K, Ballester G, Valle AC, Timo-Iaria C (1998) Studying the interaction between brain structures via directed coherence and Granger causality. Appl Signal Process 5:40–48

Baccala L, Sameshima K, Takahashi DY (2007) Generalized partial directed coherence. In: Proceedings of the 2007 15th international conference on digital signal processing pp 163–166

Baselli G, Cerutti S, Badilini F, Biancardi L, Porta A, Pagani M, Lombardi F, Rimoldi O, Furlan R, Malliani A (1994) Model for the assessment of heart period and arterial pressure variability interactions and of respiration influences. Med Biol Eng Comput 32: 143–152

Baselli G, Porta A, Rimoldi O, Pagani M, Cerutti S (1997) Spectral decomposition in multichannel recordings based on multivariate parametric identification. IEEE Trans Biomed Eng 44:1092–1101

Bernasconi C, Konig P (1999) On the directionality of cortical interactions studied by structural analysis of electrophysiological recordings. Biol Cybern 81:199–210

Brovelli A, Ding M, Ledberg A, Chen Y, Nakamura R, Bressler SL (2004) Beta oscillations in a large-scale sensorimotor cortical network: directional influences revealed by Granger causality. Proc Natl Acad Sci USA 101:9849–9854

Chatfield C (1977) Some recent developments in time-series analysis. J R Stat Soc Ser A 140:492–510

Cox DR (1981) Statistical analysis of time series: some recent developments. Scand J Stat 8:93–115

Ding M, Bressler SL, Yang W, Liang H (2000) Short-window spectral analysis of cortical event-related potentials by adaptive multivariate autoregressive modeling: data preprocessing, model validation, and variability assessment. Biol Cybern 83:35–45

Eichler M (2005) A graphical approach for evaluating effective connectivity in neural systems. Philos Trans Roy Soc B 360:953–967

Erla S, Faes L, Tranquillini E, Orrico D, Nollo G (2009) Multivariate autoregressive model with instantaneous effects to improve brain connectivity estimation. Int J Bioelectromag 11(2):74–79

Faes L, Porta A, Cucino R, Cerutti S, Antolini R, Nollo G (2004) Causal transfer function analysis to describe closed loop interactions between cardiovascular and cardiorespiratory variability signals. Biol Cybern 90:390–399

Faes L, Widesott L, Del Greco M, Antolini R, Nollo G (2006) Causal cross-spectral analysis of heart rate and blood pressure variability for describing the impairment of the cardiovascular control in neurally mediated syncope. IEEE Trans Biomed Eng 53:65–73

Faes L, Porta A, Nollo G (2009) Surrogate data approaches to assess the significance of directed coherence: application to EEG activity propagation. 31st Ann Int Conf IEEE Eng Med Biol Soc EMBC2009, pp 6280–6283

Geweke J (1982) Measurement of linear dependence and feedback between multiple time series. J Am Stat Assoc 77:304–313

Gourevitch B, Bouquin-Jeannes RL, Faucon G (2006) Linear and nonlinear causality between signals: methods, examples and neurophysiological applications. Biol Cybern 95:349–369

Granger CWJ (1969) Investigating causal relations by econometric models and cross-spectral methods. Econometrica 37:424–438

Ito J, Nikolaev AR, van Leeuwen C (2005) Spatial and temporal structure of phase synchronization of spontaneous alpha EEG activity. Biol Cybern 92:54–60

Kaminski M, Blinowska K, Szelenberger W (1995) Investigation of coherence structure and EEG activity propagation during sleep. Acta Neurobiol Exp (Wars) 55:213–219

Kaminski M, Blinowska K, Szelenberger W (1997) Topographic analysis of coherence and propagation of EEG activity during sleep and wakefulness. Electroencephalogr Clin Neurophysiol 102:216–227

Kaminski M, Ding M, Truccolo WA, Bressler SL (2001) Evaluating causal relations in neural systems: granger causality, directed transfer function and statistical assessment of significance. Biol Cybern 85:145–157

Kay SM (1988) Modern spectral estimation. Theory & application. Prentice Hall, Englewood Cliffs, NJ

Koopmans LH (1974) The spectral analysis of time series. Academic Press, New York

Korhonen I, Mainardi L, Loula P, Carrault G, Baselli G, Bianchi A (1996) Linear multivariate models for physiological signal analysis: theory. Comput Methods Programs Biomed 51:85–94

Li WK (2004) Diagnostic checks in time series. Chapman and Hall/CRC, Boca Raton, NY

Lutkepohl H (1993) Introduction to multiple time series analysis. Springer, Berlin

Malliani A, Pagani M, Lombardi F, Cerutti S (1991) Cardiovascular neural regulation explored in the frequency domain. Circulation 84:482–492

Mullen TJ, Appel ML, Mukkamala R, Mathias JM, Cohen RJ (1997) System identification of closed loop cardiovascular control: effects of posture and autonomic blockade. Am J Physiol 272:H448–H461

Nollo G, Faes L, Porta A, Antolini R, Ravelli F (2005) Exploring directionality in spontaneous heart period and systolic pressure variability interactions in humans. Implications in baroreflex gain evaluation. Am J Physiol Heart Circ Physiol 288:H1777–H1785

Pereda E, Quiroga RQ, Bhattacharya J (2005) Nonlinear multivariate analysis of neurophysiological signals. Prog Neurobiol 77:1–37

Pierce DA (1979) R-squared measures for time series. J Am Stat Assoc 74:901–910

Pierce DA, Haugh LD (1977) Causality in temporal systems. J Econom 5:265–293

Porta A, Furlan R, Rimoldi O, Pagani M, Malliani A, van de Borne P (2002) Quantifying the strength of the linear causal coupling in closed loop interacting cardiovascular variability signals. Biol Cybern 86:241–251

Porta A, Aletti F, Vallais F, Baselli G (2009) Multimodal signal processing for the analysis of cardiovascular variability. Philos Transact A Math Phys Eng Sci 367:391–409

Sato JR, Takahashi DY, Arcuri SM, Sameshima K, Morettin PA, Baccala LA (2009) Frequency domain connectivity identification: an application of partial directed coherence in fMRI. Hum Brain Mapp 30:452–461

Saul JP, Berger RD, Chen MH, Cohen RJ (1989) Transfer function analysis of autonomic regulation. II. Respiratory sinus arrhythmia. Am J Physiol 256:H153–H161

Schelter B, Winterhalder M, Eichler M, Peifer M, Hellwig B, Guschlbauer B, Lucking CH, Dahlhaus R, Timmer J (2006) Testing for directed influences among neural signals using partial directed coherence. J Neurosci Methods 152:210–219

Schlogl A, Supp G (2006) Analyzing event-related EEG data with multivariate autoregressive parameters. Prog Brain Res 159: 135–147

Shimizu S, Hoyer PO, Hyvarinen A, Kerminen A (2006) A linear non-Gaussian acyclic model for causal discovery. J Mach Learn Res 7:2003–2030

Wehling S, Simion C, Shimojo S, Bhattacharya J (2007) Assessment of connectivity patterns from multivariate time series by partial directed coherence. Chaos Compl Lett 2:413–433

Yang H, Chang JY, Woodward DJ, Baccala LA, Han JS, Luo F (2005) Coding of peripheral electrical stimulation frequency in thalamocortical pathways. Exp Neurol 196:138–152